# 07 · Case 4, part 2: breathing and blood pH

The words of this track, as it says them. Bold brown is the examiner.

**Exam case 4, continued. Breathing and blood pH.**

One idea runs through this whole track, so get it first. Carbon dioxide and hydrogen ions move together. Whatever happens to one happens to the other. And more hydrogen ions means a lower pH.

**One mark. What is normal blood pH?**

Seven point three five to seven point four five, averaging seven point four. Below seven point three five is acidotic. Above seven point four five is alkalotic.

**One mark. Write the chemical equation, in words or in formulas, for the bicarbonate buffer system in the blood.**

Carbon dioxide plus water, in balance with carbonic acid, in balance with hydrogen ions plus bicarbonate ions. In formulas: C O 2 plus H 2 O, to H 2 C O 3, to H plus, plus H C O 3 minus. And the arrows go both ways.

**Three marks. How do changes in the respiratory rate and depth, like in Manaia's panic attack, affect the blood pH?**

The hook: blow off carbon dioxide, hydrogen ions fall, pH rises. Alkalosis.

Changing the rate and depth of breathing changes the carbon dioxide level in the alveoli, and so in the blood. Blood carbon dioxide directly drives the bicarbonate buffer equation, so the same change happens to the hydrogen ion level. And a change in hydrogen ions is a change in pH.

So in Manaia's hyperventilation, she blows off carbon dioxide. The buffer uses up hydrogen ions and bicarbonate to replace it. Hydrogen ions fall, so the pH rises. That's respiratory alkalosis. Hypoventilation, after a head injury for example, does the opposite: carbon dioxide up, hydrogen ions up, pH down. Respiratory acidosis.

**Two marks. How does the respiratory system respond when something causes the blood pH to change? Think back to ketoacidosis in B N 1.**

Now it's the other way round: the pH problem comes first, and breathing fixes it.

Metabolic acidosis, with hydrogen ions up, like in ketoacidosis. The body responds with hyperventilation. Hydrogen ions are converted into carbon dioxide and water, and the extra carbon dioxide is breathed off. The breathing rate increases.

Metabolic alkalosis, with hydrogen ions down. The body responds with hypoventilation. Carbon dioxide is kept in the body, and turned into hydrogen ions and bicarbonate, to replace the ones that were lost. The breathing rate decreases.

The same buffer works both ways. Too acid, breathe faster. Too alkaline, breathe slower.

**Once more. Breathing and blood pH.**

Normal blood pH is seven point three five to seven point four five. The buffer: carbon dioxide plus water, carbonic acid, hydrogen ions plus bicarbonate, both ways. Breathing sets the blood's carbon dioxide, and hydrogen ions follow it, so pH follows too. Manaia's hyperventilation blows off carbon dioxide, hydrogen ions fall, and pH rises: respiratory alkalosis. Hypoventilation is the reverse: respiratory acidosis. In metabolic acidosis the body hyperventilates, to breathe off the extra carbon dioxide; in metabolic alkalosis it hypoventilates, to keep it.